

The diagram shows the wiring between an ESP32-C6-DEVKIT (M1A) and a stepper motor driver module. The ESP32 is connected to the driver's 3V3 pins via two 3V3 power pins. The I2C pins (SDA and SCL) are connected to the driver's I2C pins. The driver has several control pins: PUMP_MAIN, STEP_PH_DN, STEP_NUT_A, STEP_NUT_B, STEP_PDIS, and STEP_BUS. The ESP32's pins are connected to these as follows: I010 to PUMP_MAIN, I011 to STEP_PH_DN, I015 to STEP_NUT_A, I019 to STEP_NUT_B, I020 to STEP_PDIS, I012 to RST, I013 to D+/I013, I014 to D-/I012, I017 to RX0/I017, I016 to TX0/I016, I021 to RX1/I021, I022 to TX1/I022, and I023 to RTS1/I023. The driver also has a 1k resistor connected to the STEP_BUS pin.

12C SENSORS

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pH Sensor

The pH sensor circuit uses the ATLAS_EZO_pH module (M2) connected to a PR3 BL_pH sensor. The module's VCC, GND, RX/SDA, and TX/SCL pins are connected to the system. The RX/SDA line is connected to the SDA1 pin of the ADM3260 (U3) I2C-to-serial converter. The TX/SCL line is connected to the SCL1 pin of the ADM3260. The ADM3260 is powered by a 3V3 supply through a network of capacitors (C15, C20, C16, C21) and resistors (R13, R15, R17, R19). The ADM3260's SDA2 and SCL2 pins are connected to the I2C_SDA and I2C_SCL pins of the ATLAS_EZO_pH module. The module's I2C_SDA and I2C_SCL pins are connected to the I2C_SDA and I2C_SCL pins of the ATLAS_EZO_pH module. The module's I2C_SDA and I2C_SCL pins are connected to the I2C_SDA and I2C_SCL pins of the ATLAS_EZO_pH module.

Electro Conductivity Sensor

The Electro Conductivity sensor circuit uses the ATLAS_EZO_EC module (M3) connected to a PR4 BLEC sensor. The module's VCC, GND, RX/SDA, and TX/SCL pins are connected to the system. The RX/SDA line is connected to the SDA1 pin of the ADM3260 (U4) I2C-to-serial converter. The TX/SCL line is connected to the SCL1 pin of the ADM3260. The ADM3260 is powered by a 3V3 supply through a network of capacitors (C17, C22, C18, C23) and resistors (R14, R16, R18, R20). The ADM3260's SDA2 and SCL2 pins are connected to the I2C_SDA and I2C_SCL pins of the ATLAS_EZO_EC module. The module's I2C_SDA and I2C_SCL pins are connected to the I2C_SDA and I2C_SCL pins of the ATLAS_EZO_EC module. The module's I2C_SDA and I2C_SCL pins are connected to the I2C_SDA and I2C_SCL pins of the ATLAS_EZO_EC module.

Water Temperature Sensor

The Water Temperature sensor circuit uses the ATLAS_EZO_RTD module (M4) connected to a PR5 ATLAS_RTD sensor. The module's VCC, GND, RX/SDA, and TX/SCL pins are connected to the system. The RX/SDA line is connected to the I2C_SDA pin of the ATLAS_EZO_RTD module. The TX/SCL line is connected to the I2C_SCL pin of the ATLAS_EZO_RTD module. The module's I2C_SDA and I2C_SCL pins are connected to the I2C_SDA and I2C_SCL pins of the ATLAS_EZO_RTD module. The module's I2C_SDA and I2C_SCL pins are connected to the I2C_SDA and I2C_SCL pins of the ATLAS_EZO_RTD module.

Light Sensor (optional)

The optional Light sensor circuit uses the H2 JST_PH module connected to a PR2 LIGHT sensor. The module's VCC, GND, RX/SDA, and TX/SCL pins are connected to the system. The RX/SDA line is connected to the I2C_SDA pin of the H2 JST_PH module. The TX/SCL line is connected to the I2C_SCL pin of the H2 JST_PH module. The module's I2C_SDA and I2C_SCL pins are connected to the I2C_SDA and I2C_SCL pins of the H2 JST_PH module. The module's I2C_SDA and I2C_SCL pins are connected to the I2C_SDA and I2C_SCL pins of the H2 JST_PH module.

WATER LEVEL SENSOR/SWITCHES

MAIN PUMP & VALVE DRIVERS

Main Pump

24V

D3 SS34 Flyback

H3 MC_0.2"

P4 DC40F-2460

P3 MC_0.2"

T3 IRLR2905

T2 BT3904

R8 100Ω

R9 100Ω

R10 10k

C12 220μF/50V

C9 100nF

PUMP_MAIN

FLOAT_HIGH

Stay off during SPI reset

ATO Valve

24V

D4 SS34

H5 MC_0.15"

V1 DIGITEN-24V

P6 MC_0.15"

T5 AO3400A

T4 BT3904

R21 100Ω

R22 4k7

R23 10k

C30 100nF

ATO_VALVE

FLOAT_LOW

TEST POINTS

The diagram illustrates the test points for the I2C interface. On the left, three test points are shown: TP1 (24V), TP2 (5V), and TP3 (3V3), each with an upward-pointing arrow. Below them are labels TP1, TP2, and TP3. To the right, a test point TP4 is shown with a downward-pointing arrow. Further right, two test points are shown: TP5 (I2C_SDA) and TP6 (I2C_SCL), each with a rightward-pointing arrow. Below them are labels TP5 and TP6. The labels I2C_SDA and I2C_SCL are enclosed in red boxes.

24V
5V
3V3

TP1 TP2 TP3

TP4

I2C_SDA TP5 TestPoint
I2C_SCL TP6 TestPoint

PERISTALTIC PUMP DRIVERS

Ph ↓ Pump

Ph ↓ Pump

STEP_PH_DN

STEP_BUS

STEP_DIR

STEP_PDIS

3V3

24V

U5 TMC2209

H6 JST_XH

P10 ANKO_A200SX

Nutrition Pump A

Nutrition Pump A

STEP_NUT_A

STEP_BUS

STEP_DIR

STEP_PDIS

3V3

24V

U6 TMC2209

H7 JST_XH

P11 ANKO_A200SX

Nutrition Pump B

Nutrition Pump B

STEP_NUT_B

STEP_BUS

STEP_DIR

STEP_PDIS

3V3

24V

U7 TMC2209

H8 JST_XH

P12 ANKO_A200SX